F79MA 2025-26
Assessed Project 1 Solutions

1. Introduction

This project investigates the statistical properties of the maximum likelihood estimator (MLE) for the location parameter θ of the given distribution where the pdf is

$$f(x;\theta) = \tfrac{1}{2}\exp(-|x-\theta|), \quad -\infty < x < \infty,\ \theta \in \mathbb{R},$$

with cdf

$$F(x;\theta) = \begin{cases} \tfrac{1}{2}\exp(x-\theta), & x < \theta, \\ 1 - \tfrac{1}{2}\exp(-(x-\theta)), & x \geq \theta. \end{cases}$$

This distribution is in fact a Laplace(θ,1) distribution. We derive the likelihood, log-likelihood, score function, Fisher information (definition-based), and Cramér–Rao Lower Bound (CRLB), and conduct simulations in R to evaluate bias, variance, mean squared error (MSE), and approximate confidence intervals.

We observe that the maximum likelihood estimator $\hat{\theta}$ has a relatively large bias for small sample sizes n, while its variance is also relatively large. However, as n increases, the bias decreases, and the variance approaches the Cramér–Rao Lower Bound (CRLB). Consequently, the MLE becomes asymptotically unbiased, consistent, and efficient. Therefore, this estimator is recommended for use with larger datasets.

2. Mathematical Derivations

In Section 2.1 we will derive formulae for the likelihood $L(\theta;\underline{x})$ and log-likelihood $l(\theta;\underline{x}) = \log(L(\theta;\underline{x}))$ for an observed sample $\underline{x}$ of size n from the given distribution, in Section 2.2 we will derive formulae for the score function $U(\theta)$, Fisher information function $I(\theta)$ and Cramer–Rao Lower Bound (CRLB) for a random sample $\underline{X}$ of size n, and in Section 2.3 we will derive a formula for the maximum likelihood estimator (MLE) $\hat{\theta}$ for a random sample $\underline{X}$ of size n.

2.1 The Likelihood and Log-likelihood

For a sample $\underline{x} = (x_1, \ldots, x_n)$, the likelihood of the given distribution is

$$L(\theta; \underline{x}) = \prod_{i=1}^{n} f(x_i; \theta) = \left(\frac{1}{2}\right)^n \exp\Big(- \sum_{i=1}^{n} |x_i - \theta| \Big),$$

and the log-likelihood is

$$\ell(\theta; \underline{x}) = -n \log 2 - \sum_{i=1}^{n} |x_i - \theta|.$$

Assumptions: We consider an i.i.d. random sample $\underline{x} = (x_1, \ldots, x_n)$ from the distribution

$$f(x; \theta) = \frac{1}{2}\, e^{-|x-\theta|}, \quad x \in \mathbb{R},\ \theta \in \mathbb{R}.$$

We assume standard regularity conditions, so that differentiation under the integral sign is valid and the Fisher information is finite. Although the pdf is not differentiable at $x = \theta$, this occurs with probability zero and does not affect the analysis.

2.2 The score function, Fisher information and Cramer–Rao Lower Bound

From the above, the log-likelihood $l(\theta; \underline{X})$ for a random sample $\underline{X}$ of size n is

$$\ell(\theta; \underline{X}) = -n \log 2 - \sum_{i=1}^{n} |X_i - \theta|.$$

Taking the derivative with respect to θ gives the score function $U(\theta)$ as follows.

Score function: For $X_i \neq \theta$ for all i, the score function is given by

$$U(\theta) = \frac{\partial \ell(\theta; \underline{X})}{\partial \theta} = \frac{\partial}{\partial \theta}\Big[- \sum_{i=1}^{n} |X_i - \theta| \Big] = \sum_{i=1}^{n} \operatorname{sign}(X_i - \theta),$$

where

$$\operatorname{sign}(X_i - \theta) = \begin{cases} 1, & X_i > \theta, \\ -1, & X_i < \theta, \end{cases}$$

and the score function is undefined if $X_i = \theta$ for some i.

Fisher information: For a single observation, using the definition

$$I_1(\theta) = \mathrm{E}\Big[\big(\partial_\theta \log f(X; \theta) \big)^2 \Big] = \mathrm{E}[(\operatorname{sign}(X - \theta))^2] = 1.$$

For n independent observations,

$$I_n(\theta) = n \cdot I_1(\theta) = n.$$

Cramér–Rao Lower Bound (CRLB):

$$\mathrm{Var}(\hat{\theta}) \geq \frac{1}{I_n(\theta)} = \frac{1}{n}.$$

2.3 The Maximum Likelihood Estimator

MLE using the score function
The maximum likelihood estimator (MLE) $\hat{\theta}$ for a random sample $\underline{X}$ of size n from a distribution with a smooth density function whose domain is the whole of $\mathbb{R}$ is the solution of $U(\hat{\theta}) = 0$.
For $X_i \neq \theta$ for all i, the score function is given by

$$U(\theta) = \sum_{i=1}^{n} \mathrm{sign}(X_i - \theta).$$

where

$$\mathrm{sign}(X_i - \theta) = \begin{cases} 1, & X_i > \theta, \\ -1, & X_i < \theta. \end{cases}$$

The MLE satisfies $U(\hat{\theta}) = 0$, which means that the number of observations below $\hat{\theta}$ equals the number above. Hence, the MLE is the sample median:

$$\hat{\theta}_{\text{MLE}} = \mathrm{median}(X_1, X_2, \ldots, X_n).$$

For odd n, the solution is unique (the middle order statistic). For even n, any value between the two central order statistics is an MLE.

[**Alternative solution using sum of absolute deviations**

The log-likelihood for a random sample $\underline{X}$ of size n is

$$\ell(\theta; \underline{X}) = -n \log 2 - \sum_{i=1}^{n} |X_i - \theta|.$$

Maximizing $\ell(\theta; \underline{X})$ with respect to θ is equivalent to minimizing

$$Q(\theta) = \sum_{i=1}^{n} |X_i - \theta|.$$

It is a well-known property that the value of θ that minimizes the sum of absolute deviations is the *sample median*. Therefore,

$$\hat{\theta}_{\text{MLE}} = \text{median}(X_1, \ldots, X_n).$$

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Comment on attainability of CRLB: The CRLB is not attained exactly in finite samples because the MLE (the sample median) is not differentiable at the data points and is biased for finite n. These nonsmooth points violate the usual regularity conditions required for exact efficiency. However, as $n \to \infty$, the sample median becomes asymptotically unbiased and satisfies

$$\sqrt{n}\,(\hat{\theta}_{\text{MLE}} - \theta) \xrightarrow{d} N\big(0, \tfrac{1}{n}\big)\,,$$

so the CRLB is attained asymptotically. Therefore, the MLE is *asymptotically efficient*.

3. Simulations in R

To accompany this report, we have written an R script `F79MA_Assignment1.R`. Using the seed 1234, we will simulate the given distribution in R. This gives a parameter `My_theta` of 0.1137034 and an $m \times n$ matrix `samples` consisting of $m = 1000$ simulations of samples of size $n = 100$ from the given distribution with parameter `My_theta` which we will use for our analysis.

3.1 Calculations for a single sample

Using the first row of the matrix `samples`, which corresponds to a single sample of size $n = 100$, we find

$$\hat{\theta} = 0.01489375 \quad (\texttt{Quantity1})$$

and

$$l(\hat{\theta}; \underline{x}) = -185.7264 \quad (\texttt{Quantity2}).$$

In Figure 1, we have plotted the log-likelihood function. This demonstrates that `Quantity1` (vertical red line) is indeed the maximum likelihood estimate, as the log-likelihood at this point (`Quantity2`, red point) is the maximum of the function. The plot also indicates that, for this particular sample, the MLE is noticeably different from the true parameter value `My_theta` (vertical blue line), suggesting that the estimator may exhibit non-negligible bias and variance in finite samples.

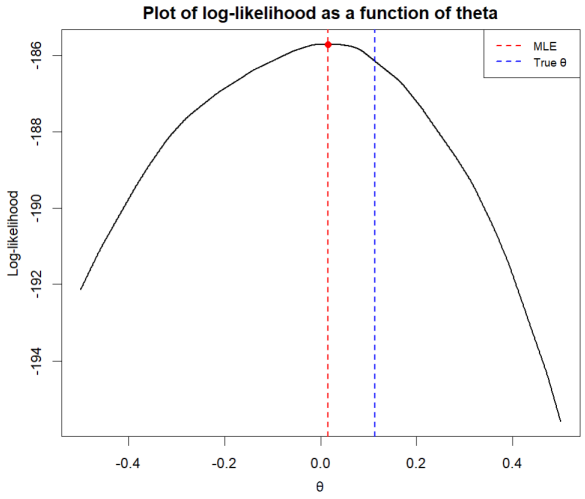


Figure 1: Plot of log-likelihood with MLE and true parameter value indicated with red and blue lines.

Confidence Interval using Asymptotic Normality

For a large sample size n, the maximum likelihood estimator $\hat{\theta}$ is approximately normally distributed:

$$\hat{\theta} \sim \mathrm{N}\Big(\theta, \frac{1}{n}\Big),$$

since the Cramér–Rao Lower Bound (CRLB) is $1/n$. Using $n = 100$, the approximate 90% confidence interval for θ is

$$\hat{\theta} \pm z_{0.95}\sqrt{\frac{1}{n}} = 0.01489375 \pm 1.6449\sqrt{\frac{1}{100}} = \big(-0.1496,\ 0.1794\big),$$

where the lower and upper limits are saved in R as `Quantity3` and `Quantity4`, respectively.

Confidence Interval using Deviance (Likelihood-Ratio)

Asymptotically, the deviance statistic

$$D(\theta) = 2\Big(l(\hat{\theta}; \underline{x}) - l(\theta; \underline{x})\Big)$$

follows approximately a χ^2_1 distribution. Using this result, an approximate 90% confidence interval for θ is

$$\big(-0.2246,\ 0.1912\big),$$

where the lower and upper limits are stored in R as `Quantity5` and `Quantity6`, respectively.

Comment: We observe that both confidence intervals include the true θ. The asymptotic normality CI is narrower and symmetric, whereas the deviance-based CI is wider and asymmetric, better capturing the shape of the log-likelihood function.

3.2 Calculations based on many samples

We now make use of the whole matrix `samples` to estimate the bias, variance and mean squared error (MSE) of the maximum likelihood estimator by calculating the maximum likelihood estimate corresponding to each row of the matrix.

The bias $\mathbf{E}\,(\hat{\theta} - \theta)$ can be estimated as

$$\frac{1}{m}\sum_{j=1}^{m}(\hat{\theta}_j - \theta) = 0.00337441 \quad (\texttt{Quantity7}).$$

The variance $\mathbf{Var}\,(\hat{\theta})$ can be estimated as

$$\frac{1}{m-1}\sum_{j=1}^{m}(\hat{\theta}_j - \bar{\theta})^2 = 0.01179492 \quad (\texttt{Quantity8}).$$

The MSE can be estimated as

$$\frac{1}{m}\sum_{j=1}^{m}(\hat{\theta}_j - \theta)^2 = 0.01179451 \quad (\texttt{Quantity9}).$$

For $n = 100$, the CRLB is $1/100 = 0.01$ (`Quantity10`), which we see is close to `Quantity8` above.

Figure 2 shows the histogram of MLEs calculated from the matrix `samples` with the density of $\mathrm{N}(\theta, 1/n)$ added. We see that the asymptotic normality assumption we used in Section 3.1 seems reasonable.

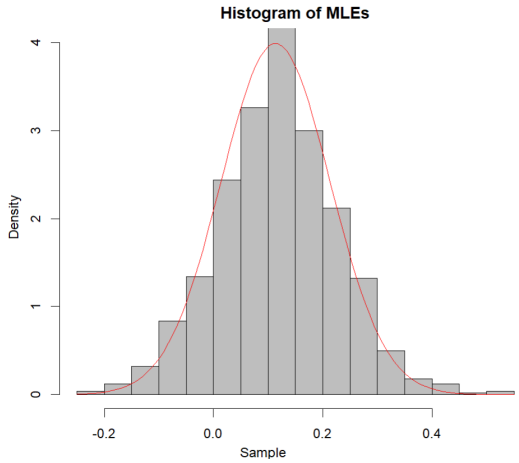


Figure 2: Histogram of MLE values with asymptotic normal density curve added in red

3.3 Bias, variance and MSE as functions of sample size

We now plot how the bias, variance and MSE vary as the sample size n increases from $n = 3$ up to $n = 100$.

Figures 3, 4, and 5 show the bias, variance, and MSE of the MLE as functions of n. A curve representing the CRLB has been added to Figure 4, closely matching the variance even for small n. While all three measures approach 0 as n increases, the bias decreases more slowly, albeit with small absolute values, whereas the variance decreases faster but starts from relatively larger values. These trends indicate that the MLE is consistent, with both bias and variance diminishing as the sample size grows, and that it becomes increasingly efficient, approaching the Cramér–Rao Lower Bound for larger n. The MSE follows a similar pattern to the variance.

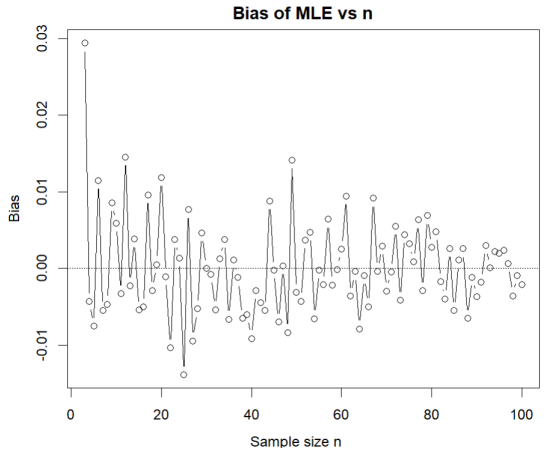


Figure 3: Plot of bias as a function of sample size n

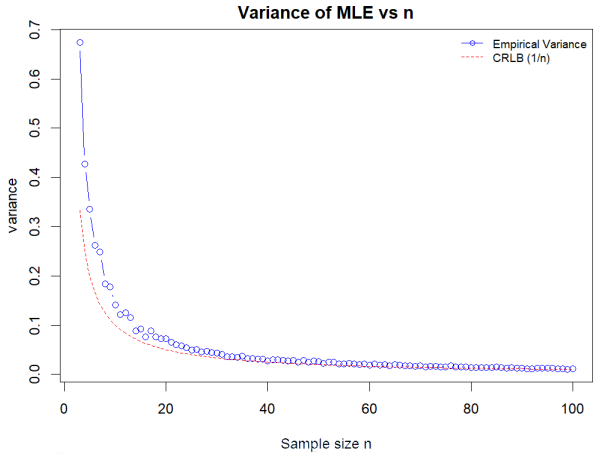


Figure 4: Plot of variance as a function of sample size n with CRLB added as the red dashed line

Consistency and efficiency: The MLE is consistent, with bias and variance decreasing as the sample size increases, and is asymptotically efficient, approaching the CRLB for large n. For smaller n, the variance is slightly above the CRLB, but efficiency improves as n grows.

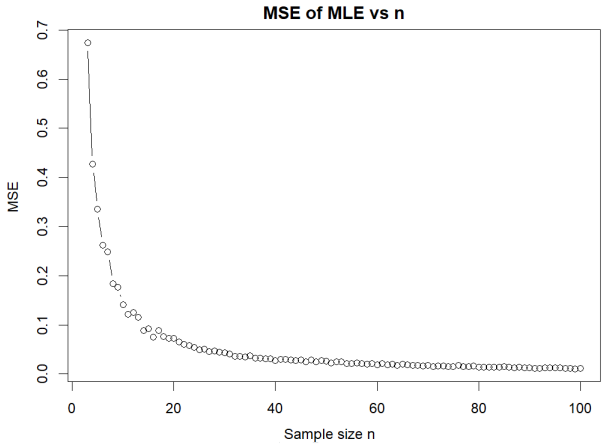


Figure 5: Plot of MSE as a function of sample size n

4. Conclusion

In this report, we studied the maximum likelihood estimator (MLE) $\hat{\theta}$ for a particular distribution. The estimator exhibits a small bias that decreases as the sample size increases, with bias values mostly within ± 0.02. The variance is relatively high for small sample sizes but decreases with n, approaching the Cramér–Rao Lower Bound (CRLB) of $1/n$. For $n = 100$, the simulated variance (0.0118) is close to the CRLB (0.01), indicating that the MLE becomes more precise as the sample size grows.

The mean squared error (MSE), which combines bias and variance, also decreases with increasing n, confirming the good estimation properties of the MLE. These patterns indicate that the MLE is both consistent and asymptotically efficient, performing increasingly well for larger samples.

For the given distribution (Laplace), the MLE, given by the sample median, is simple and performs well in simulations. Despite the Fisher information not being strictly defined due to non-differentiability at data points, the MLE is *consistent* and *asymptotically normal*, making it suitable in practice, especially for heavy-tailed risks. We recommend its use in models with moderate to large datasets and suggest investigating other estimators if smaller bias or variance is desired.